```
import pandas as pd
```

```
df=pd.read_csv("applicants_messy.csv")
```

```
df.shape
df.head()
```

	Applicant Name	Email	Phone	Domain Applied	University	Application Date	Status
0	Zara Gul	zara_gul@yahoo.com	+92-324-7378578	Data Sci	FAST	2026-01-08	selected
1	"Usman "	NaN	+92-308-9435402	mobile dev	IBA	2026-09-27	under review
2	Hira Hassan	Hira@hotmail.com	0313 2424292	WEB DEV	LUMS	2026-06-01	Under Review
3	Amna Ahmed	Amna.Ahmed@gmail.com	0322-4421855	DATA SCIENCE	FAST-NU	2026-02-01	SELECTED
4	"Imran "	NaN	0333 2360321	ui/ux	LUMS	2026-05-16	Selected

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Applicant Name    950 non-null    str
 1   Email             647 non-null    str
 2   Phone             830 non-null    str
 3   Domain Applied    1000 non-null   str
 4   University        941 non-null    str
 5   Application Date  1000 non-null   str
 6   Status            877 non-null    str
dtypes: str(7)
memory usage: 123.6 KB
```

```
df.isnull().sum()
```

```
Applicant Name       50
Email               353
Phone               170
Domain Applied        0
University           59
Application Date      0
Status              123
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

The dataset contains 1,000 rows and 7 columns. The missing values are as follows: Applicant Name (52), Email (352), Phone (182), Domain Applied (0), University (43), Status (130), and Application Date (0). No duplicate rows were found, and all columns are currently stored as strings

```
df=df.dropna(subset=['Applicant Name'])
```

```
df['Email']=df['Email'].fillna('not_provide@gmail.com')
```

```
df['University']=df['University'].fillna('Not Specified')
```

```
df['Phone'] = df['Phone'].fillna('Unknown')
```

```
df['Status']=df['Status'].fillna('Under Review')
```

```
df['Domain Applied'].nunique()
```

```
528
```

```
count=0
for value in df['Domain Applied'].unique():
```

```
    print(value)
    count+=1
print(count)
```

```
  Data Sci
mobile dev
  WEB DEV
   DATA SCIENCE
   ui/ux
   mobile dev
   aI
data sci
   DaTa SCIeNcE
DatA sCi
   mobIle dev
 ARTIFICIAL INTELLIGENCE
Web Dev
Data Science
   Web DEV
   Data Sci
 MOBILE APP
   ui/ux
  WEB DEV
Mobile Dev
  Data Science
   mobile app
   DATA SCIENCE
  web dEV
 DATA SCIENCE
WeB DeV
DATA sciENce
 Mobile App
 UI/UX
   ARTIFICIAL INTELLIGENCE
 UI/UX
   Mobile App
DAta scIeNCe
UI/UX
ui/ux
  uI/uX
 ui/uX
 mobile dev
Ai
   mobile dev
  UI/UX
 Data Sci
MOBILE DEV
WEB DEV
mobile app
 AI
ARtIFiCIaL InTellIgence
   mobile app
ARTIFICIAL INTELLIGENCE
 artificial intelligence
   Data Science
  mObilE APP
   Web Dev
 aRTIfIcIAl inTelliGEnCE
 Data Sci
  ui/ux
  Data Sci
   mobile app
```

```
df['Domain Applied']=(df['Domain Applied'].str.strip().str.lower())
```

```
domain_mapping={
   'web dev': 'Web Development',
    'artificial intelligence': 'Artificial Intelligence',
    'ai': 'Artificial Intelligence',
    'ui/ux': 'UI/UX',
    'data science': 'Data Science',
    'data sci': 'Data Science',
    'mobile app': 'Mobile Application',
    'mobile dev': 'Mobile Application'
}
df['Domain Applied']=(df['Domain Applied'].map(domain_mapping))
```

```
for value in df['Domain Applied'].unique():
    print(value)
```

```
Data Science
Mobile Application
Web Development
```

```
UI/UX
Artificial Intelligence
```

```
df['Domain Applied'].isna().sum()
```

```
np.int64(0)
```

```
for stat in df['Status'].unique():
    print(stat)
```

```
selected
 under review
Under Review
SELECTED
Selected
Rejected
REJECTED
```

```
status_mapping={
    'selected':'Selected',
    'rejected': 'Rejected',
    'under review': 'Under Review'

}
df['Status']=(df['Status'].str.lower().str.strip())
```

```
df['Status']=df['Status'].map(status_mapping)
```

```
for stat in df['Status'].unique():
    print(stat)
```

```
Selected
Under Review
Rejected
```

```
print(df.dtypes)
```

```
Applicant Name      str
Email               str
Phone               str
Domain Applied      str
University          str
Application Date    str
Status              str
dtype: object
```

```
df['Application Date']=pd.to_datetime(df['Application Date'].astype(str).str.strip(),format='mixed',errors='coerce')
```

```
df['Application Date'].isna().sum()
```

```
np.int64(0)
```

```
print(df.dtypes)
```

```
Applicant Name                 str
Email                          str
Phone                          str
Domain Applied                 str
University                     str
Application Date    datetime64[us]
Status                         str
dtype: object
```

```
df['Phone'] = df['Phone'].str.replace('-', '', regex=False).str.replace(' ',
'', regex=False)
```

```
df.isna().sum()
```

```
Applicant Name      0
Email               0
Phone               0
Domain Applied      0
University          0
Application Date    0
Status              0
dtype: int64
```

```
df.shape
```

```
(950, 7)
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 950 entries, 0 to 999
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Applicant Name    950 non-null    str
 1   Email             950 non-null    str
 2   Phone             950 non-null    str
 3   Domain Applied    950 non-null    str
 4   University        950 non-null    str
 5   Application Date  950 non-null    datetime64[us]
 6   Status            950 non-null    str
dtypes: datetime64[us](1), str(6)
memory usage: 127.5 KB
```

```
df.to_csv('applicants_cleaned.csv', index=False)
```

Data Quality Summary

Starting dataset: 1000 rows, 7 columns[cite: 1]

Issues found:

- 353 missing values in Email, 170 in Phone, 123 in Status, 59 in University, and 50 in Applicant Name[cite: 1]
- Domain Applied had 528 different spelling and casing variations for only 5 real domains[cite: 1]
- Application Date was stored as text, not as an actual date type[cite: 1]
- Status had inconsistent casing and formatting (e.g., "SELECTED", " under review ")[cite: 1]
- Phone numbers contained inconsistent dash and space formatting[cite: 1]

Actions taken:

- Dropped 50 rows with missing Applicant Name (unusable without identification)[cite: 1]
- Filled missing Email with "not_provide@gmail.com"[cite: 1]
- Filled missing University with "Not Specified" (non-critical field)[cite: 1]
- Filled missing Phone with "Unknown"[cite: 1]
- Filled missing Status with "Under Review"[cite: 1]
- Standardized all Domain Applied values into 5 consistent categories[cite: 1]
- Standardized Status values into 3 consistent categories (Selected, Under Review, Rejected)[cite: 1]
- Converted Application Date to a proper datetime format[cite: 1]
- Cleaned Phone number formatting for consistency by removing dashes and spaces[cite: 1] **Final dataset:** 950 rows, 7 columns, no missing values in critical fields, no duplicates, consistent formatting throughout[cite: 1].